

EPA: Easy Prompt Augmentation on Large Language Models via Multiple Sources and Multiple Targets

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Abstract

Large language models (LLMs) have shown promising performance on various NLP tasks via task prompting. And their performance can be further improved by appending task demonstrations to the head of the prompt. And usually, a better performance can be achieved with more demonstrations. However, asking the users to write the demonstrations can be cumbersome. As a simple yet cost-effective workaround, this paper proposes a novel method called EPA (Easy Prompt Augmentation)¹ that effectively minimizes user efforts in writing demonstrations while improving the model performance at the same time. EPA achieves these goals by automatically augmenting the demonstrations with multiple sources/targets, where each of them paraphrases each other. This is well motivated as augmenting data via paraphrasing effectively improves neural language models. EPA thus employs paraphrasing as an augmentation method for in-context learning. Extensive experiments indicate that EPA effectively improves both NLU and NLG tasks, covering from natural language inference to machine translation in translating tens of languages.²

1 Introduction

Large language models (LLMs) possess the ability to carry out various understanding and generation tasks from natural language inference to machine translation (Brown et al., 2020; Lin et al., 2022; Zhang et al., 2022; Wei et al., 2022; Wang et al., 2023). Such an ability is closely related to in-context learning (Rubin et al., 2022; Zhang et al., 2022). In-context learning prepends one or more source/target pairs (namely demonstrations) to the

head of the requests. This effectively improves the downstream task performance. However, most of the scientific research constraints to the situation where those demonstrations are always available (Min et al., 2022; Dong et al., 2022).

Yet, the above-mentioned situation is unrealistic, as human annotation is expensive, and there is no guarantee that there always exist enough demonstrations. This is an important consideration, especially for commercial products, where we would like to reduce user efforts in writing demonstrations. This subsequently leads to a better user experience.

In contrast, this paper considers a realistic situation where only a few demonstrations are available. We propose a simple yet effective framework called EPA (Easy Prompt Augmentation) that creates multiple sources and multiple targets respectively where they paraphrase each other.

EPA is also well-motivated by the prior works (Gao et al., 2020; Lu and Lam, 2023) that doing data augmentation via paraphrasing improves the neural language model during the model training stage. EPA considers paraphrasing both sources and targets to enhance in-context learning. In summary, this paper makes three key contributions:

- We propose a novel method that is easy to use called EPA, aiming at improving the performance of LLMs via in-context learning.
- EPA yields promising performance on various NLP tasks, covering from natural language inference to machine translation.
- In-depth analyses indicate that EPA brings good improvement, and naively copying the demonstration degrades the performance.

EPA is a simple yet effective method for improving LLMs. Therefore, we hope that EPA can benefit our community not only in terms of research but also in developing commercial products.

¹While this paper considers augmenting prompts via demonstrations, we name it EPA as the name EDA is already taken by a well-known NLP method (Wei and Zou, 2019).

²Code and data will be released upon publication.

2 Easy Prompt Augmentation

Motivation EPA is inspired by the fact that paraphrasing can improve the evaluation of natural language generation tasks by enriching the target side of the test instances (Thompson and Post, 2020; Bawden et al., 2020; Tang et al., 2023). This is due to the fact that one meaning can usually be represented in several sentences in terms of natural language, and multiple targets can provide a precise evaluation of the actual meaning of the generations.

Meanwhile, the rise of LLMs provides a unified solution to different language-related tasks via the interface of prompts. Prompting can be further enhanced by in-context learning, where source/target pairs can be prepended to the head of the prompts to serve as demonstrations. Yet, the construction of demonstrations usually requires human efforts, which can be clumsy. This is especially inconvenient for commercial products, where we would like to reduce user efforts as much as possible.

Our Approach Motivated by the need to automatically construct demonstrations instead of human efforts, this paper proposes and investigates the effectiveness of paraphrasing when a small set of demonstrations are already available. Specifically, EPA assumes there are a small set of demonstrations that are already. EPA then considers paraphrasing the demonstrations on both **source side** and **target side** to create new demonstrations for in-context learning. Formally, assuming we have one demonstration, the traditional in-context learning feeds the following text concatenation into LLMs:

$$[x_d, y_d, x],$$

where x_d and y_d represent the source and target sentence of the demonstration pair. x represents the actual prompt that we hope LLMs solve. EPA first paraphrases x_d and y_d into additional demonstrations $x_{d1}, x_{d2}, x_{d3}, \dots, x_{dn}$ paired with $y_{d1}, y_{d2}, y_{d3}, \dots, y_{dn}$ where x s and y s represents the same meaning. n represents the number of additional paraphrases. EPA then feeds the following text concatenation into LLMs:

$$[x_d, y_d, x_{d2}, y_{d2}, x_{d3}, y_{d3}, \dots, x_{dn}, y_{dn}, x]$$

The whole EPA framework aims at improving the performance of LLMs while minimizing human efforts in writing demonstrations simultaneously. Therefore, EPA can reduce user efforts and improve the user experience for commercial products.

3 Experimental Setup

3.1 Datasets

We conduct extensive experiments to verify the effectiveness of EPA on a variety of both understanding and generation tasks:

- **Machine Translation** We randomly select 45 high-resourced and low-resourced languages from FLORES-200 (NLLB-Team, 2022), which is an MT dataset that contains about 200 languages with 1,012 parallel sentences included in the dataset, which were extracted from English Wikipedia covering a variety of topics and domains. The sentences were curated manually by professional translators from English into other languages. We report on translating all the languages in FLORES-200 from English, with the complete dev-test that has about 1,000 instances.
- **Dialogue Summarization** SAMSum (Gliwa et al., 2019) is a dialogue summarization dataset created manually created by linguists who are fluent in English. We use the test set that contains 819 instances for evaluations.
- **Paraphrasing** Quora Question Pairs (QQP) is a paraphrasing dataset that requires generating an alternative surface form in the same language that maintains the original semantic meaning. We use the preprocessed version from Gong et al. (2022) that contains about 4.65k test instances.
- **Natural Language Inference** The SNLI corpus (Bowman et al., 2015) is a collection of manually-written English sentence pairs labelled for balanced classification between entailment, contradiction, and neutral relationship. We use the test set that contains about 10k instances. The MNLI corpus (Williams et al., 2018) is a collection of manually-written sentence pairs annotated with textual entailment information. It is modelled based on the SNLI corpus. MNLI covers a range of categories of spoken and written text. MNLI also supports a distinctive evaluation of cross-genre generalization. We use the matched validation set with about 9.82k instances.

3.2 Evaluation Metrics

Machine Translation For the evaluation on MT, we report chrF++ scores (Popović, 2015) com-

Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA
acm_Arab	23.97	26.28	afr_Latn	51.97	70.16	als_Latn	59.41	63.04	arz_Arab	24.84	29.38	asm_Beng	12.87	27.03
azb_Arab	1.81	2.84	bak_Cyrl	18.99	39.03	bjn_Arab	2.28	2.38	crh_Latn	26.38	27.69	dan_Latn	57.85	64.02
eus_Latn	45.86	47.10	ewe_Latn	28.11	40.42	fin_Latn	67.78	75.32	fuv_Latn	12.24	18.16	gaz_Latn	30.76	36.62
guj_Gujr	23.12	41.61	jpn_Jpan	30.00	37.57	kas_Deva	1.58	9.09	khk_Cyrl	37.40	65.40	khm_Khmr	24.17	28.47
kik_Latn	20.44	35.82	kir_Cyrl	21.24	36.73	kmb_Latn	8.78	21.23	kor_Hang	20.54	27.69	lij_Latn	45.80	48.41
lim_Latn	27.70	28.23	lmo_Latn	16.79	17.86	ltg_Latn	22.93	27.63	lug_Latn	24.31	29.76	mai_Deva	21.82	51.40
mya_Mymr	25.23	32.03	nld_Latn	53.61	55.63	nya_Latn	20.34	59.73	oci_Latn	53.78	54.19	ory_Orya	30.57	32.71
pag_Latn	28.99	41.78	scn_Latn	20.87	24.27	slv_Latn	39.19	51.92	sna_Latn	42.16	47.41	snd_Arab	25.15	36.20
szl_Latn	33.62	34.22	tpi_Latn	39.00	45.93	tsn_Latn	28.13	56.48	tum_Latn	39.36	60.29	vie_Latn	43.75	52.84

Table 1: Comparison between GPT-3.5-TURBO and EPA. Results in chrF++ for MT on the FLORES-200 dataset. The best results are bolded and highlighted. We report on translating from English into the languages (3-shot on GPT).

Language	Copy-9	EPA	Language	Copy-9	EPA	Language	Copy-9	EPA	Language	Copy-9	EPA	Language	Copy-9	EPA
acm_Arab	15.99	26.28	aeb_Arab	21.12	30.02	apc_Arab	30.38	33.26	bjn_Arab	1.77	2.38	dan_Latn	56.59	64.02
dyu_Latn	9.07	17.77	dyu_Latn	9.07	17.77	dzo_Tibt	2.44	34.06	fij_Latn	28.31	31.76	fin_Latn	59.49	75.32
hat_Latn	45.88	56.81	hat_Latn	45.88	56.81	isl_Latn	56.26	59.90	kaz_Cyrl	52.09	58.58	kea_Latn	32.23	34.07

Table 2: Comparison between Copy-9 and EPA. Results in chrF++ for MT on the FLORES-200 dataset. The best results are bolded and highlighted. We report on translating from English into the languages (3-shot on EPA).

puted by the sacreBLEU repository.³ We also use COMET (Bosselut et al., 2019) with the model version eamt22-cometinno-da.

Dialogue Summarization, Paraphrasing For the evaluation on dialogue summarization and paraphrasing, we use F1 scores on ROUGE-L (Lin, 2004) and BLEU.

Natural Language Inference We use accuracy to evaluate the performance on the task of NLI.

3.3 Baselines and Prompt Design

ChatGPT We experiment with ChatGPT, a multilingual large language model that has shown strong abilities across various NLU and NLG tasks (Wang et al., 2023). At the time of writing, this LLM is widely popular. We use the ChatGPT model versioned GPT-3.5-TURBO. We access ChatGPT via the official API through Python.

Copy-9 As one of the baselines, we repeat the original three demonstrations three times to get 9 demonstrations, which is equal in the number of demonstrations to EPA.

Prompt Design We use the following prompts for each task we conducted in our experiments:

- **Machine Translation** *Translate the following text from English into [target-lang]: [source]*
- **Dialogue Summarization** *Given the following dialogue: [source] Give the dialogue summarization:*
- **Paraphrasing** *Given the English input: [source] Give the paraphrase:*

³<https://github.com/mjpost/sacrebleu>

- **Natural Language Inference** *Given the following two sentences: [source] Whether they are neutral, contradiction, or entailment:*

For the dataset of FLORES-200, we use three pairs of demonstrations for in-context learning on our baselines. For the remaining datasets except for paraphrasing, we use one or three pairs of demonstrations on our baselines. We randomly select the demonstrations from the training sets. We use one demonstration for paraphrasing. All results except for MT are averaged from three runs on three different sets of demonstrations.

3.4 Easy Prompt Augmentation

We use ChatGPT to create the paraphrases based on the randomly selected demonstrations. We use the following prompt to create the paraphrases: *Paraphrase the following text: [source]*. For FLORES-200, we create three paraphrases for each English instance and we use the NLLB translator⁴ to acquire the paraphrases in the target language, making it multiple sources and multiple targets. For the remaining tasks, we create one paraphrase for each pair of demonstrations using ChatGPT solely. EPA uses the newly created paraphrases as additional demonstrations for in-context learning.

4 Results

4.1 Main Results

Machine Translation Table 1 presents the results in chrF++ on 45 languages for translating from English into these languages. These lan-

⁴<https://huggingface.co/spaces/Narrativaai/NLLB-Translator>

Model	BLEU	ROUGE-L
<i>1-shot</i>		
GPT	9.10 $\pm$ 0.21	38.22 $\pm$ 0.25
EPA	9.51 $\pm$ 0.16	38.98 $\pm$ 0.14
<i>3-shot</i>		
GPT	9.51 $\pm$ 0.18	38.77 $\pm$ 0.19
EPA	9.89 $\pm$ 0.12	39.48 $\pm$ 0.27

Table 3: Evaluations of EPA and GPT on dialogue summarization. The upper half for 3-shot demonstrations on GPT and EPA applies paraphrasing to the demonstrations. The lower half for 5-shot demonstrations on GPT and EPA applies paraphrasing to the demonstrations. Results averaged from three sets of demonstrations. We found EPA always improve GPT on each set of demonstration.

Model	BLEU	ROUGE-L
GPT	11.22 $\pm$ 0.09	28.90 $\pm$ 0.12
EPA	11.44 $\pm$ 0.07	29.24 $\pm$ 0.06

Table 4: Evaluations of EPA and GPT on paraphrasing. Results averaged from three sets of demonstrations. We found EPA always improve GPT on each set of demonstration.

Model	SNLI-Accuracy	MNLI-Accuracy
<i>1-shot</i>		
GPT	54.94 $\pm$ 0.56	53.37 $\pm$ 0.38
EPA	57.48 $\pm$ 0.43	55.14 $\pm$ 0.29
<i>3-shot</i>		
GPT	54.80 $\pm$ 0.44	50.00 $\pm$ 0.42
EPA	57.57 $\pm$ 0.48	55.01 $\pm$ 0.52

Table 5: Evaluations of EPA and GPT on natural language inference. The upper half for 3-shot demonstrations on GPT and EPA applies paraphrasing to the demonstrations. The lower half for 5-shot demonstrations on GPT and EPA applies paraphrasing to the demonstrations. Results averaged from three sets of demonstrations. We found EPA always improve GPT on each set of demonstration.

guages range from high-resource languages to low-resource languages. We compare ChatGPT (GPT-3.5-TURBO) with EPA. The results indicate that EPA brings clear improvements. The gains can be large, by up to about 6x chrF++ points on low-resource languages (1.58 to 9.09 for English to Kashmiri written in Devanagari script (kas_Deva)) and by up to about 3x chrF++ points on high-resource languages (20.34 to 59.3 for English to Nyanja written in Latin script (nya_Latn)). The improvement is also consistent for different languages written in different scripts. The results in Table 6 in the Appendix also show the same trend. EPA reports an average score of -0.557, which clearly exceeds the score of -0.994 reported by the baseline. This indicates that EPA is an effective approach for improving in-context learning on LLMs.

Dialogue Summarization Table 3 presents the results in BLEU and ROUGE-L on GPT and EPA, where EPA shows good improvement with up to

0.79 improvements in F1 scores on ROUGE-L.

Paraphrasing Table 4 presents the results in BLEU and ROUGE-L on GPT and EPA, where EPA shows good improvement compared to GPT.

Natural Language Inference Table 5 reports an evaluation of SNLI and MNLI on the task of NLI. We observe a large performance gain with EPA, by up to 5.01 accuracy on MNLI with 3-shot in-context learning. This indicates the effectiveness of EPA that enhances in-context learning on NLU tasks with LLMs.

4.2 Analysis - Copy Demonstration

Table 2 reports chrF++ scores on 15 randomly selected languages from NLLB-200 with Copy-9. We observe that copying the demonstrations does not surpass EPA. The performance difference can be large, by up to about 15x difference (2.44 to 34.06 for English to Dzongkha written in Tibetan script (dzo_Tibt)). We postulate that copying the demonstration can lead to ‘overfitting’ on in-context learning and makes the prompting less generalizable. In contrast, EPA creates paraphrasings in different surface forms and hence improves the generalization of the in-context demonstration learning.

5 Related Work and Conclusions

In-context learning effectively improves downstream task performance. However, most of the scientific research constraints to the situation where those demonstrations are always available (Min et al., 2022; Dong et al., 2022). This then limits the practical use of in-context learning. Meanwhile, prior works (Gao et al., 2020; Lu and Lam, 2023) have shown that conducting paraphrasing for data augmentation effectively improves the neural language model during the model training stage. Combining these motivations, we propose a novel method called EPA (Easy Prompt Augmentation) that effectively reduces user efforts in writing demonstrations while improving the model performance simultaneously. EPA achieves these goals by augmenting demonstrations with multiple sources/targets, where each of them paraphrases each other. Extensive experiments indicate that EPA effectively improves both NLU and NLG tasks, covering from natural language inference to machine translation in translating tens of languages. The in-depth analysis also indicates that EPA surpasses naively copying the demonstrations.

Limitations

This paper presents an analysis of around 40 languages only on MT. However, there are more than thousands of languages around the world. We leave more investigations to our future work. EPA requires an automatic paraphraser to be effective. We do not perform an investigation on the situation where human paraphraser are available. EPA is also less useful when there exist many demonstrations to be used. Finally, we also conduct experimentation on ChatGPT, which could affect reproducibility in the future.

Ethical Statement

We honour and support the ACL Code of Ethics. There is no ethical issue known to us in this work. A well-known and widely used LLM is used in our work, which is subjected to generating offensive context. Yet the above-mentioned issues are widely known to commonly exist for LLMs. Any content generated does not reflect the view of the authors.

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A COMET Scores

Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA	Language	GPT	EPA
acm_Arab	-0.96	-0.22	afr_Latn	-1.13	-0.52	als_Latn	-0.96	-0.71	arz_Arab	-1.14	0.05	asm_Beng	-1.07	-0.23
azb_Arab	-1.04	-0.74	bak_Cyrl	-0.99	-0.08	bjn_Arab	-1.13	0.68	crh_Latn	-0.76	-0.51	dan_Latn	-0.98	-0.49
eus_Latn	-0.83	-0.31	ewe_Latn	-1.13	-0.34	fin_Latn	-0.84	-0.39	fuv_Latn	-0.71	-0.43	gaz_Latn	-0.97	-0.35
guj_Gujr	-0.86	0.42	jpn_Jpan	-0.52	-0.12	kas_Deva	-1.11	-0.13	khk_Cyrl	-0.95	0.43	khm_Khmr	-1.11	-0.84
kik_Latn	-0.99	-0.32	kir_Cyrl	-1.36	-0.42	kmb_Latn	-0.63	-0.29	kor_Hang	-0.88	-0.39	lij_Latn	-0.70	-0.33
lim_Latn	-1.02	-0.17	lmo_Latn	-1.14	-0.27	ltg_Latn	-1.11	-0.29	lug_Latn	-1.37	-0.59	mai_Deva	-1.01	-0.43
mya_Mymr	-0.99	-0.55	nld_Latn	-1.10	0.42	nya_Latn	-1.05	-0.13	oci_Latn	-1.10	0.25	ory_Orya	-0.83	-0.50
pag_Latn	-1.32	0.20	scn_Latn	-1.14	-0.31	slv_Latn	-1.02	-0.48	sna_Latn	-0.91	-0.32	snd_Arab	-0.95	-0.17
szl_Latn	-0.84	-0.91	tpi_Latn	-1.11	-0.05	tsn_Latn	-1.04	-0.28	tum_Latn	-1.07	-0.25	vie_Latn	-0.99	0.02

Table 6: Comparison between GPT-3.5-TURBO and EPA. Results in COMET for MT on the FLORES-200 dataset. The best results are bolded and highlighted. We report on translating from English into the languages (3-shot on GPT).